

CLASS 02 / SEP 1

NON-DESTRUCTIVE MODELING

+ ASSIGNMENT 0.1 ISSUED TODAY

MODULE 0 / PROCEDURAL / ARC 3133

TUTORIAL 0.1 — THE LOGIC

Not the buttons. The four things a graph actually does.

01

A GRAPH IS A CHAIN OF OPERATIONS

Each node takes something in, does one thing, passes it on.

02

DATA FLOWS LEFT TO RIGHT

If a result is wrong, walk backwards until it stops being wrong.

03

GEOMETRY IS AN OUTPUT, NOT A THING

Nothing is 'made'. The graph describes how to make it, every time.

04

EVERY INPUT STAYS LIVE

Change a number at the start and everything downstream recomputes.

WHAT BROKE

Bring the file, not the description. Let's fix it live.

NOTHING APPEARS

Usually a preview toggled off, or a node not connected to output.

THE WRONG NUMBER OF THINGS

A list where a single item was expected, or the reverse.

IT WORKED, THEN STOPPED

Something upstream changed. Walk backwards node by node.

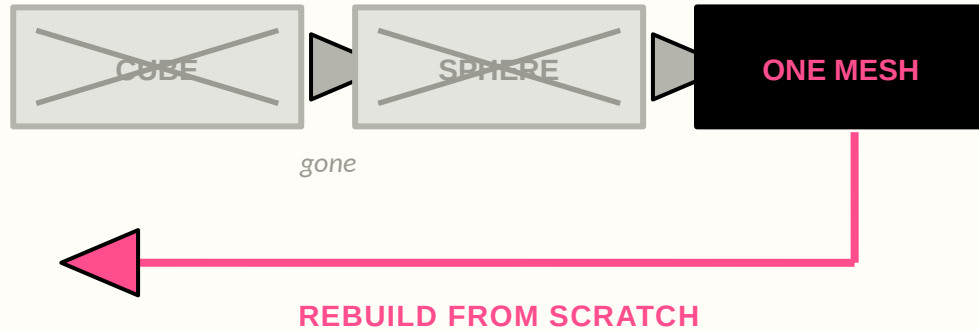
A NODE IS ORANGE / RED

It is telling you what it needs. Read the message before changing anything.

WHY NON-DESTRUCTIVE

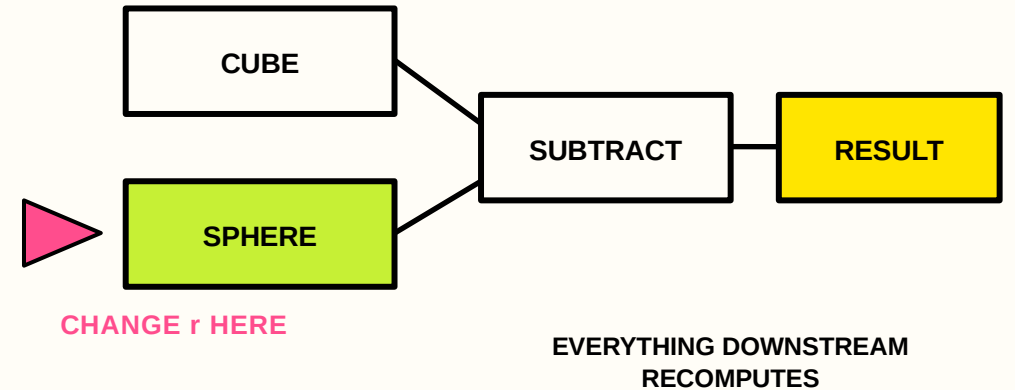
Same job both times: make the sphere bigger. Watch what it costs.

DESTRUCTIVE



The operations were discarded. The sphere no longer exists as anything you can edit.

NON-DESTRUCTIVE

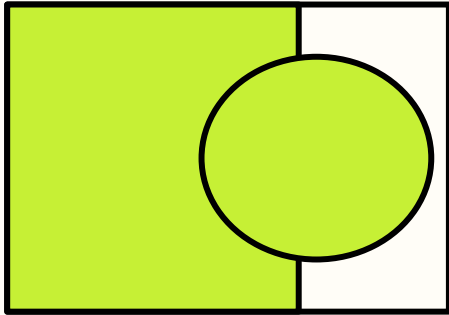


Every input is still a live node. Edit it and the result rebuilds itself.

0.1 REQUIRES YOUR OPERATION HISTORY TO BE INTACT. I WILL ASK TO SEE IT.

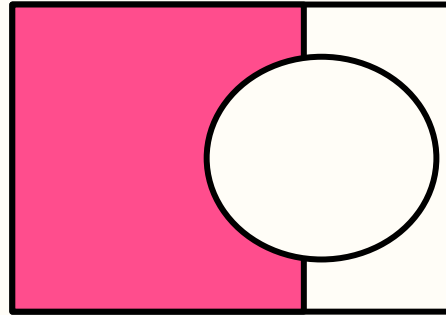
THREE BOOLEANS

Everything in Module 0 is built from these, applied in order.



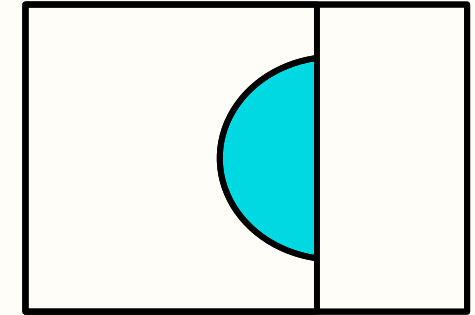
UNION

$A + B$



DIFFERENCE

$A - B$



INTERSECTION

$A \cap B$

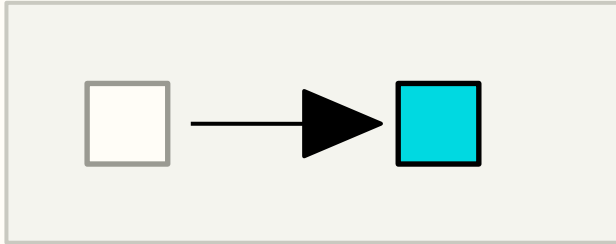
Order matters. $A - B$ is not $B - A$. Your instructions must make the order unambiguous.

TRANSFORMATIONS ARE ARITHMETIC

Three operations on a coordinate. Nothing more mysterious than that.

MOVE

= ADD



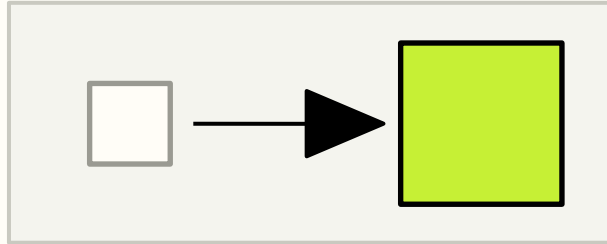
THE MATH

$$\begin{aligned}x' &= x + dx \\y' &= y + dy \\z' &= z + dz\end{aligned}$$

$$(2, 1) + (3, 0) \rightarrow (5, 1)$$

SCALE

= MULTIPLY



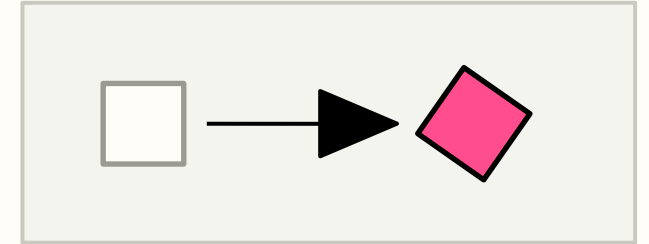
THE MATH

$$\begin{aligned}x' &= x \times f \\y' &= y \times f \\z' &= z \times f\end{aligned}$$

$$(2, 1) \times 2 \rightarrow (4, 2)$$

ROTATE

= MIX x AND y



THE MATH

$$\begin{aligned}x' &= x \cdot \cos\theta - y \cdot \sin\theta \\y' &= x \cdot \sin\theta + y \cdot \cos\theta\end{aligned}$$

$$(1, 0) \text{ rot } 90^\circ \rightarrow (0, 1)$$

Move and scale treat each coordinate on its own. Rotation is the only one where x and y have to talk to each other.

PRECEDENT — THE MASSING DIAGRAM

One drawing per operation. The sequence is the argument.



Add: a sequential massing diagram, one axo per move.

WHAT MAKES IT WORK

- One operation per frame. Never two.
- The same viewpoint every frame, so the change is the only thing moving.
- The moved or removed part is marked — colour, arrow, ghost.
- It reads left to right without a caption.

Your diagram is graded on whether the sequence can be read — not on how the building looks.

ISSUED TODAY

0.1 — PROCEDURE: DIAGRAM + INSTRUCTIONS

DUE CLASS 4 — SEP 15 / 8%

Take an existing building. Rebuild its massing as a sequence of primitives and Booleans. Then document that sequence twice.

01 THE DIAGRAM

A sequential axonometric set — one drawing per operation — plus one larger final isometric with human scale, context, shadows and annotation.

02 THE INSTRUCTIONS

A written procedure another person could follow to rebuild your object without ever seeing your model.

BOTH ON ONE 17" × 11" SHEET, IN YOUR VISUAL IDENTITY.

REQUIREMENTS

Counting these earns a C. The rest is judgement.

5+

Boolean operations

4+

distinct primitive types

2+

translations

1+

rotation

1

precedent analysis — 2–3 sentences on why the massing reads as a sequence

Operation history intact and demonstrable. No imported or downloaded models.

THE NOTATION

Write the procedure as pseudocode. Formal enough to be unambiguous, plain enough to read aloud.

PRIMITIVES

`box(x, y, z)`

`cylinder(r, h)`

`sphere(r)`

`cone(r, h)`

TRANSFORMS

`move(s, x, y, z)`

`rotate(s, axis, deg)`

`scale(s, factor)`

BOOLEANS

`union(a, b)`

`subtract(a, b)`

`intersect(a, b)`

THREE RULES

- Every shape is created centred on the origin (0, 0, 0). Move it from there.
- Units are abstract. No millimetres, no metres — 3 units, half the width.
- Every step assigns a name, so the order of a subtraction is never in doubt.

This is real: OpenSCAD is a working language built on exactly these operations.

WORKED EXAMPLE

Four operations. Read it, then draw what it makes.

PROCEDURE

```
base    = box(6, 4, 3)
tower   = box(2, 2, 6)
tower   = move(tower, 2, 0, 3)
void     = cylinder(r=1, h=5)
void     = move(void, -1.5, 0, 0)

mass     = union(base, tower)
result  = subtract(mass, void)
```

WHAT THIS FIXES

ORDER

subtract(mass, void) can only mean one thing.

POSITION

Every shape starts at the origin, so a move is the whole story.

REUSE

A named step can be referred to later — mass is used, not rebuilt.

READING

Anyone can execute it. That is the test at Class 5.

Your sheet carries BOTH: the axonometric sequence, and the procedure that produces it.

CHOOSING YOUR OBJECT

Pick badly and the rest of the assignment is impossible.

ORTHOGONAL MASSING

Boxes, slabs, stacked volumes. Right angles.

NO COMPLEX CURVES

If you cannot get there with primitives, it is the wrong building.

5–10 CLEAR MOVES

Too simple is boring. Too complex will not resolve in five Booleans.

APPROVED OBJECT LIST

Add: 6–8 buildings that work. Students may propose others for approval.

WHAT HAPPENS TO THIS WORK

Three things, and they are not all at once.

CLASS 4

SEP 15

Sheet due — diagram and instructions.

CLASS 5

SEP 22

Instructions exchanged. You build a classmate's object from their written procedure alone. The gap between the two results is the finding.

CLASS 8

OCT 13

Your object, 3D printed at small scale, presented at the Midterm. No supports — orientation is your only lever.

NOW — BUILD IT

Rest of the session: start your operation tree with help in the room.



Object chosen and approved



First three primitives placed



At least one Boolean working



Operation history intact — nothing collapsed to a mesh

LEAVE WITH A WORKING TREE.

BEFORE NEXT CLASS

WATCH

Tutorial 0.2 — 3D printing. Manifold geometry, orientation, supports, layer height, slicing. Sent tonight.

CHECK

FabLab Safety Orientation must be complete by the end of today. Without it you cannot use any machine.

BUILD

Keep going on 0.1. Bring your tree next week — we prepare prints in class.

WRITE

Draft your instructions. Read them aloud to someone who has not seen the model.